CMPUT 654: Mathematical Foundations of Modern AI Systems Fall 2026

Lecture 1: Transformers, pretraining, inference, and a first hardness result

September 1, 2026

Lecturer: Csaba Szepesvári *Scribe: instructor model note*

Purpose of this note. This is an example of the expected scribe-note style: identify the lecture's question, define the mathematical objects, state claims with their scope, and reconstruct the central calculation. It is an exposition, not a transcript.

Delivery note. Administrative material occupied a substantial part of the meeting. The lecture reached transformer mechanics, part of the pretraining construction, autoregressive inference, and the beginning of the hardness argument. The remaining parts of the planned opening map continue on September 3.

1.1 A language model as a conditional probability model

Let the vocabulary have size V, and identify token $x_t \in [V]$ with the standard basis vector $e_{x_t} \in \{0,1\}^V$. Include a beginning-of-sequence token x_0. For a length-T token sequence, define the shifted input matrix $Z \in \{0,1\}^{T\times V}$ by

$$Z_{t,:} = e_{x_{t-1}}^\top, \qquad t = 1,\dots,T. \tag{1.1}$$

An embedding matrix $E \in \mathbb{R}^{V\times d}$ and a position matrix $P \in \mathbb{R}^{T\times d}$ give

$$H^0 = ZE + P, \qquad H^0_{t,:} = E_{x_{t-1},:} + P_{t,:}. \tag{1.2}$$

The one-hot vector is conceptual: software implements $e_{x_{t-1}}^\top E$ as a row lookup. The position term is necessary because attention acting on a set of content vectors does not by itself identify their order. Additive learned positions, fixed sinusoidal positions, and rotary positions are different ways to supply this information.

1.2 One fully specified decoder block

Suppose $H^\ell \in \mathbb{R}^{T\times d}$ enters layer ℓ. We describe a standard pre-normalized decoder block with h attention heads of width d_h. For head $a \in [h]$, let

$$U = \mathrm{LN}_{\ell,1}(H^\ell), \tag{1.3}$$

$$Q_a = UW^Q_{\ell,a}, \qquad K_a = UW^K_{\ell,a}, \qquad V_a = UW^V_{\ell,a}, \tag{1.4}$$

$$A_a = \mathrm{softmax}_{\mathrm{row}}\left(\frac{Q_aK_a^\top}{\sqrt{d_h}} + M\right), \qquad O_a = A_aV_a, \tag{1.5}$$

where

$$W^Q_{\ell,a}, W^K_{\ell,a}, W^V_{\ell,a} \in \mathbb{R}^{d\times d_h}, \qquad M_{ij} = \begin{cases} 0, & j \le i, \\ -\infty, & j > i. \end{cases} \tag{1.6}$$

The rowwise softmax makes every row of A_a a probability vector. The causal mask sets the weight of every future position to zero. Concatenate the head outputs and apply the first residual connection:

$$G = H^\ell + [O_1 \mid \cdots \mid O_h]W_\ell^O, \qquad W_\ell^O \in \mathbb{R}^{hd_h \times d}. \tag{1.7}$$

The feed-forward sublayer acts on each position separately:

$$R = \mathrm{LN}_{\ell,2}(G), \tag{1.8}$$

$$H^{\ell+1} = G + \phi\Big(RW_\ell^1 + \mathbf{1}(b_\ell^1)^\top\Big)\, W_\ell^2 + \mathbf{1}(b_\ell^2)^\top, \tag{1.9}$$

where

$$W_\ell^1 \in \mathbb{R}^{d\times d_{\mathrm{ff}}}, \quad W_\ell^2 \in \mathbb{R}^{d_{\mathrm{ff}}\times d}, \quad b_\ell^1 \in \mathbb{R}^{d_{\mathrm{ff}}}, \quad b_\ell^2 \in \mathbb{R}^{d}. \tag{1.10}$$

For a concrete pedagogical choice, $\phi(z) = z\Phi(z)$ coordinatewise is the GELU activation, with Φ the standard normal distribution function. Layer normalization acts on a row $r \in \mathbb{R}^d$ as

$$\mathrm{LN}_{\gamma,\beta}(r) = \gamma \odot \frac{r - \bar{r}\mathbf{1}}{\sqrt{d^{-1}\|r - \bar{r}\mathbf{1}\|_2^2 + \varepsilon}} + \beta. \tag{1.11}$$

Practical models change some of these choices, for example replacing LayerNorm by RMSNorm, additive positions by RoPE, or the plain MLP by a gated MLP. The equations above specify one complete decoder block; they are not a claim that this particular variant is uniquely responsible for modern performance.

1.3 The single-token view and pretraining

After L decoder blocks, the causal mask ensures that row $h_t^L = H_{t,:}^L$ depends only on the prefix $x_{<t}$. With $W_U \in \mathbb{R}^{d\times V}$ and $b_U \in \mathbb{R}^V$, define logits and the next-token distribution by

$$z_t = h_t^L W_U + b_U, \tag{1.12}$$

$$p_\theta(x_t = v \mid x_{<t}) = \frac{\exp(z_{t,v})}{\sum_{u=1}^{V} \exp(z_{t,u})}. \tag{1.13}$$

The loss on the observed token is

$$\ell_t(\theta) = -\log p_\theta(x_t \mid x_{<t}) = -z_{t,x_t} + \log \sum_{u=1}^{V} e^{z_{t,u}}. \tag{1.14}$$

Weight tying sets $W_U = E^\top$; it is common, but the probability model does not require it. Zooming back out gives the autoregressive factorization

$$p_\theta(x_{1:T}) = \prod_{t=1}^{T} p_\theta(x_t \mid x_{<t}) \tag{1.15}$$

and the empirical pretraining objective

$$\widehat{L}_{\mathrm{pre}}(\theta) = \frac{1}{N_{\mathrm{tok}}} \sum_{i,t} -\log p_\theta(x_{i,t} \mid x_{i,<t}). \tag{1.16}$$

During training, *teacher forcing* supplies the true prefix at every position. Consequently, all token losses in a sequence can be evaluated in parallel, and the same parameters receive gradients from many positions and many sequences. Schematically, an optimizer produces updates

$$\theta_{k+1} = \theta_k - \eta_k \widehat{g}_k. \tag{1.17}$$

This specifies the objective and the computational graph. It does not yet explain why minimizing the objective produces reusable representations or reasoning-like computation; that is a later question in the course.

1.4 Inference is a separate algorithmic choice

At generation time the true next token is unavailable. A sampled decoder uses

$$X_t \sim p_\theta(\cdot \mid X_{<t}), \tag{1.18}$$

and feeds the sampled token back into the next step. Greedy decoding instead uses

$$X_t \in \arg\max_v p_\theta(v \mid X_{<t}). \tag{1.19}$$

These procedures are attached to the same trained conditional distribution but produce different sequence distributions. In particular, greedy decoding need not find a maximum-probability complete sequence: the locally best first token can lead to prefixes whose continuations all have small probability.

Exercise 1.1 (Question raised in lecture)**.** Construct a family of length-T conditional distributions for which greedy decoding returns a sequence whose probability is exponentially smaller in T than that of the maximum-probability sequence. Explain why one-token optimality of an argmax decision does not justify repeated argmax under a sequence-level objective.

1.5 A first hardness result: unseen arbitrary labels

The first mathematical question was:

When can examples determine behavior on inputs absent from the data?

Let $\mathcal{X}$ be finite and let P be a distribution on $\mathcal{X}$. Draw a target $F : \mathcal{X} \to \{0, 1\}$ uniformly from all $2^{|\mathcal{X}|}$ labelings. Let

$$S = ((X_i, F(X_i)))_{i=1}^m, \qquad X_i \overset{\text{iid}}{\sim} P, \tag{1.20}$$

and let A be any learner, possibly randomized. For two predictors define

$$R_P(f, g) = \mathbb{E}_{X\sim P}[\mathbb{I}\{f(X) \neq g(X)\}]. \tag{1.21}$$

Theorem 1.2 (Missing-mass lower bound)**.** *For every learner A,*

$$\mathbb{E}\big[R_P(A(S), F)\big] \geq \frac{1}{2} \sum_{x\in\mathcal{X}} P(x)(1 - P(x))^m. \tag{1.22}$$

Proof idea. At a point absent from the sample, the corresponding random label remains a fair bit even after conditioning on everything the learner observed. The probability that x is absent from m independent draws is $(1 - P(x))^m$; summing half of this missing mass gives the bound.

Proof. For $x \in \mathcal{X}$, let $E_x = \{x \notin \{X_1, \ldots, X_m\}\}$. Then

$$\mathbb{E}[R_P(A(S), F)] = \sum_{x \in \mathcal{X}} P(x)\mathbb{P}(A(S)(x) \neq F(x)) \tag{1.23}$$

$$\geq \sum_{x \in \mathcal{X}} P(x)\mathbb{P}(A(S)(x) \neq F(x), E_x). \tag{1.24}$$

On E_x, the sample contains no information about the independent fair bit $F(x)$. Hence

$$\mathbb{P}(A(S)(x) \neq F(x), E_x) = \frac{1}{2}\mathbb{P}(E_x). \tag{1.25}$$

Also,

$$\mathbb{P}(E_x) = \prod_{i=1}^{m} \mathbb{P}(X_i \neq x) = (1 - P(x))^m. \tag{1.26}$$

Substitution gives (1.22). □

Exercise 1.3 (The formal independence step)**.** Prove (1.25) formally, including a randomized learner. One route is to condition on the sample inputs, all labels $F(z)$ for $z \neq x$, and the learner's independent random seed.

Corollary 1.4 (Uniform domain)**.** *If P is uniform on N points, then*

$$\mathbb{E}[R_P(A(S), F)] \geq \frac{1}{2}\left(1 - \frac{1}{N}\right)^m. \tag{1.27}$$

The random target is a proof device, not a claim about natural tasks. Averaging over all labelings also gives a deterministic statement.

Corollary 1.5 (A fixed hard target exists)**.** *For every learner A, at least one fixed labeling $f : \mathcal{X} \to \{0, 1\}$ satisfies*

$$\mathbb{E}_{S_f, A}[R_P(A(S_f), f)] \geq \frac{1}{2} \sum_{x \in \mathcal{X}} P(x)(1 - P(x))^m, \tag{1.28}$$

where $S_f = ((X_i, f(X_i)))_{i=1}^m$.

Proof. Let $g(f) = \mathbb{E}_{S_f, A}[R_P(A(S_f), f)]$. The theorem lower-bounds the average of $g(F)$ over uniformly randomized labelings. At least one fixed labeling has $g(f)$ at least as large as that average. □

This proof pattern is the *randomization hammer*: randomize the target, prove that every learner has large average error, and extract one deterministic hard target. The result says that arbitrary unseen labels require coverage. It does not say that structured targets are unlearnable. The positive threshold contrast, where one example constrains many unseen points, was deferred to the continuation of Lecture 1.

1.6 Questions carried forward

Two concrete questions from the meeting will be developed in homework.

1. **Can position be encoded relatively?** For sinusoidal encodings, a shift is a block rotation of the encoding. RoPE applies those rotations to queries and keys so that their dot product contains a rotation indexed by the relative displacement. The essential calculation is

$$(R(s)q_s)^\top(R(t)k_t) = q_s^\top R(t-s)k_t.$$

   The homework will specify the rotation matrices and ask what this identity does and does not imply.

2. **What is wrong with tokenwise argmax?** It is optimal for a single categorical decision under zero-one loss. Repeating it is generally neither sampling from the model nor global optimization of a sequence-level objective. A homework construction will make the gap exponential.

1.7 Take-home messages

- A decoder-only transformer defines every next-token conditional through a fully specified sequence of embeddings, masked attention, residual blocks, and a softmax readout.
- Pretraining evaluates log loss at all positions under true prefixes; generation feeds model-produced tokens back sequentially.
- The learned probability model and the decoding algorithm are distinct objects.
- No learner can infer independent arbitrary labels at unseen points. The lower bound identifies missing task information, not a universal failure of learning on structured problems.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. Gomez, L. Kaiser, and I. Polosukhin. Attention Is All You Need. *NeurIPS*, 2017. https://arxiv.org/abs/1706.03762.

[2] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu. RoFormer: Enhanced Transformer with Rotary Position Embedding. 2021. https://arxiv.org/abs/2104.09864.